

St. Jude's General Hospital - Oncology Service
DISCHARGE SUMMARY

PATIENT: MARTIN, Susan

MRN: AML011

DOB: 23-Jun-1966

ADMISSION DATE: 15-Nov-2023

DISCHARGE DATE: 19-Mar-2024

ADMITTING DIAGNOSIS: Newly Diagnosed Acute Myeloid Leukemia.

DISCHARGE DIAGNOSIS: Acute Myeloid Leukemia with inv(16)(p13.1q22), Favorable-Risk by ELN 2022 Criteria, in Complete Remission.

CONSULTING SERVICES:

- Infectious Disease
- Cardiology
- Clinical Pharmacy
- Nutrition

ATTENDING PHYSICIAN: Dr. Eleanor Vance, MD

SUMMARY OF HOSPITAL COURSE:

REASON FOR ADMISSION:

Ms. Susan Martin is a 57-year-old female with a recent history of progressively worsening fatigue, ecchymosis, and a single episode of epistaxis lasting 20 minutes. Her past medical history is notable for well-controlled Type 2 Diabetes Mellitus (on Metformin 1000 mg BID) and hypertension (on Lisinopril 10 mg daily). She presented to the emergency department on November 10, 2023, where a Complete Blood Count (CBC) revealed pancytopenia with circulating blasts.

- **Initial Labs:** WBC 2.1 K/uL (with 68% blasts), Hemoglobin 7.9 g/dL, Platelets 18 K/uL.

She was admitted directly to the hematology-oncology service for an expedited workup.

HOSPITAL COURSE:

A bone marrow biopsy performed on Day 1 of admission (11-Nov-2023) confirmed the diagnosis of Acute Myeloid Leukemia with 72% myeloblasts. Cytogenetic analysis returned on 15-Nov-2023, revealing an inversion on chromosome 16, specifically **inv(16)(p13.1q22)**, and molecular studies were negative for FLT3, IDH1/2, and NPM1 mutations. This places her into the **ELN 2022 Favorable-risk** category.

After extensive counseling regarding risks, benefits, and treatment alternatives, Ms. Martin consented to undergo intensive induction chemotherapy. A double-lumen PICC line was placed. Pre-treatment evaluation by Cardiology confirmed adequate cardiac function (ECHO showing LVEF of 60-65%) to proceed.

On **17-Nov-2023**, Ms. Martin began induction chemotherapy with the standard **"7+3" regimen**, consisting of a continuous infusion of Cytarabine for seven days and bolus infusions of Daunorubicin on days 1, 2, and 3.

Her hospital course was complicated by the expected period of profound myelosuppression.

- **Neutropenic Fever:** On Day +9 post-chemotherapy (26-Nov-2023), she developed a fever to 38.9°C with an Absolute Neutrophil Count (ANC) of <100 cells/uL. Blood cultures were drawn, and she was empirically started on broad-spectrum intravenous antibiotics (Cefepime). Infectious Disease was consulted. Despite a comprehensive workup, including chest X-ray and urinalysis, no clear source of infection was identified, and blood cultures remained negative. Her fevers resolved within 72 hours, and she completed a 10-day course of antibiotics.
- **Transfusion Support:** Throughout her nadir period, she required significant transfusion support. She received a total of 8 units of packed red blood cells (pRBCs) to maintain a hemoglobin >7.0 g/dL and 12 units of apheresis platelets to maintain a count >10 K/uL.
- **Mucositis:** She developed Grade 3 oral mucositis, which was managed effectively with patient-controlled analgesia (PCA) with morphine, viscous lidocaine, and nutritional support from our dietary team.

TREATMENT RESPONSE & RECOVERY:

Peripheral blood counts began to show signs of recovery around Day +25. A Day +28 bone marrow biopsy was performed (15-Dec-2023) to assess response. The results were highly encouraging:

- **Pathology Report (18-Dec-2023):**
 - **Specimen:** Bone Marrow Aspirate and Biopsy, Left Posterior Iliac Crest.
 - **Cellularity:** 40-50% (recovering).
 - **Blasts:** < 5%.
 - **Morphology:** Trilineage hematopoiesis is present and maturing.
 - **Flow Cytometry:** No immunophenotypic evidence of residual leukemia.
 - **Conclusion:** Morphologic and immunophenotypic Complete Remission (CR).

Following this result, her counts continued to improve. By early January 2024, she was transfusion-independent with a stable ANC above 1.0 K/uL. She was discharged for a brief period before being readmitted for her first of three planned cycles of consolidation chemotherapy with high-dose cytarabine (HiDAC). She has now successfully completed two cycles of HiDAC, with the second cycle finishing on 05-Mar-2024. This current admission was for management of her post-consolidation neutropenia, which resolved without complication.

CONDITION AT DISCHARGE:

Ms. Martin is in excellent spirits. She is afebrile, hemodynamically stable, and fully

ambulatory. She is tolerating a regular diet. Her PICC line is intact and flushing well. Her blood counts are safe for discharge.

- **Discharge Labs (19-Mar-2024):** WBC 4.5 K/uL, ANC 2.8 K/uL, Hemoglobin 10.1 g/dL, Platelets 155 K/uL.

DISCHARGE MEDICATIONS:

1. **Lisinopril** 10 mg PO daily (for HTN)
2. **Metformin** 1000 mg PO BID (for T2DM)
3. **Acyclovir** 400 mg PO BID (antiviral prophylaxis)
4. **Posaconazole** 300 mg PO daily (antifungal prophylaxis)
5. **Ondansetron** 8 mg PO every 8 hours as needed for nausea
6. **Calcium with Vitamin D** 1 tablet PO daily

FOLLOW-UP:

- **Leukemia Clinic:** Follow-up appointment with Dr. Vance in 1 week (26-Mar-2024) for CBC check and to schedule her third and final planned cycle of HiDAC consolidation therapy.
- **Primary Care Physician:** Follow-up with her PCP, Dr. Stevens, within 2 weeks for ongoing management of her diabetes and hypertension.

The patient and her family were provided with extensive education regarding signs and symptoms of infection, when to call the on-call physician, and the importance of medication adherence. All questions were answered. Prognosis is favorable, but long-term monitoring is essential.

Dr. Eleanor Vance, MD

Hematology-Oncology

Dictated: 19-Mar-2024

Transcribed: 19-Mar-2024